Optimal taxation of fossil fuel consumption and production given terms of trade effects and geopolitical externalities

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Abstract

How should the EU tax the production and consumption of oil, taking into account that this will affect its world market price, thereby reducing its import and bill and reducing revenue for Russia, thereby limiting its capacity for military aggression? This paper provides a general answer in a static model with a globally integrated oil market. A calibration implies that the optimal EU (self-interested policy), taking into account climate damages, terms of trade effect and geopolitical effects consists of taxing oil consumption at an ad valorem rate of 0.64 and subsidizing oil production at an ad valorem rate of 0.45.

1 Introduction

2 A general static model

2.1 The model

There is a globally integrated world market for oil. The world market price of oil is denoted by p. There is a set of players (countries or sets of countries, like the EU, acting together), denoted I, that constitutes a partition of the world. The players simultaneously set their vectors of ad valorem tax rates (x_i, y_i) on the consumption and production of oil. We will derive formulae characterizing the Nash equilibrium in this game.

Let $D_i((1+x_i)p)$ denote the demand for oil in the region i, given that the consumers there face the after-tax oil price of $(1+x_i)p$. Let $S_i((1-y_i)p)$ denote the supply of oil in the region i, given that the producers there face the after-tax oil price of $(1-y_i)p$. The world market price p is determined by the market clearing condition:

$$\sum_{i\in I} D_i((1+x_i)p) = \sum_{i\in I} S_i((1-y_i)p)$$

In the simultaneous move game where the players simultaneously choose (x_i, y_i), we can thus view $p((x_i, y_i)_{i\in I})$ as a function of all the tax rates. From now on we will view the world market price as this function but for better readability we will suppress its argument $(x_i, y_i)_{i\in I}$ in the notation. We shall denote by W_i the classical economic surplus accruing to player i:

$$W_k = \underbrace{\int_{v=(1+x_k)p}^{\infty} D_k(v)dv}_{\textbf{consumer surplus}} + \underbrace{px_k D_k((1+x_k)p)}_{\textbf{tax revenue}} + \underbrace{\int_{v=S_k^{-1}(0)}^{p} S_k(v)dv}_{\textbf{producer surplus}}$$

However, player k's overall utility is denoted by U_k and given as follows:

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$$U_k = W_k + \sum_{i \neq k} \alpha_{ki} W_i \underbrace{- \eta_k \sum_{i \in I} S_i((1-y_i)p)}_{\textbf{climate damages}}$$

Thus α_{ki} gives how player k values a dollar accruing to player i. In this paper, we draw on estimates for $\alpha_{EU,Russia}$, from Beaufils et al. 2025 arguing that money accruing to Russia substantially harms the EU by relaxing financial constraints that could limit the Russian aggression. Implicit in the formulation of U_k is that for player k it is immaterial how social surplus accrues to Russia. A general argument for this assumption is that the Russian government can adjust its tax policy to capture increases in social surplus as additional tax revenue. Combining this public finance theory assumption (which can be micro-founded in some settings as shown by Kaplow (2010)) with a specification of the Russian government's objective function that places some value on consumer surplus (e.g. as a proxy for limiting popular discontent, thus potentially just for instrumental reasons) and some value on having money for the military, one can micro-found the different versions of assumptions that we consider in this paper on how additional social surplus accruing to Russia translate into increase in military spending. For example, if there is a hard constraint on ensuring a minimal consumer surplus that the government needs to satisfy to limit public discontent then any excess social surplus will get spent on the military. In this example of a model, it is clear that social surplus is a sufficient statistic for determining Russian military aggression, consistent with our assumption (implicit in the specification of U_{EU}) that W_{Russia} is a sufficient statistic for how the EU gets affected by Russia.

The other term, $-\eta_k \sum_{i \in I} S_i((1-y_i)p)$, captures the climate damages internalised by player k.

Lemma 1. *Let us denote:*
$\epsilon_i^D := (1+x_i)p\frac{D_i'((1+x_i)p)}{D_i((1+x_i)p)}$
$\epsilon_i^S := (1-y_i)p\frac{S_i'((1-y_i)p)}{S_i((1-y_i)p)}$.

Also let $D = \sum_i D_i$ and let ϵ^D and ϵ^S denote the global aggregate price elasticity of demand and supply. We have:

$$\frac{\frac{\partial p}{\partial x_k}}{p} = \frac{1}{(1+x_k)} \frac{\epsilon_k^D \frac{D_k}{D}}{\epsilon^S - \epsilon^D}$$

$$\frac{\frac{\partial p}{d y_k}}{p} = \frac{1}{(1-y_k)} \frac{\epsilon_k^S \frac{S_k}{S}}{\epsilon^S - \epsilon^D}$$

Proof. Differentiating the market clearing condition with respect to x_k yields:

$$pD_k'((1+x_k)p) + \frac{\partial p}{\partial x_k} \sum_{i \in I}(1+x_i)D_i'((1+x_i)p) = \frac{\partial p}{\partial x_k} \sum_{i \in I}(1-y_i)S_i((1-y_i)p)$$

With our notation, we get:

$$\frac{1}{1+x_k}\epsilon_k^D D_k + \frac{\frac{\partial p}{\partial x_k}}{p}(\sum_{i \in I} \epsilon_i^D D_i - \sum_{i \in I} \epsilon_i^S S_i) = 0$$

From this we get:

$$\frac{\frac{\partial p}{\partial x_k}}{p} = \frac{1}{(1+x_k)} \frac{\epsilon_k^D \frac{D_k}{D}}{\epsilon^S - \epsilon^D}$$

Differentiating the market clearing condition

$$\sum_{i \in I} D_i((1+x_i)p) = \sum_{i \in I} S_i((1-y_i)p)$$

with respect to y_k yields:

$$\frac{\partial p}{\partial y_k} \sum_{i \in I}(1+x_i)D_i'((1+x_i)p) = \frac{\partial p}{\partial x_k} \sum_{i \in I}(1-y_i)S_i((1-y_i)p) - pS_k'((1-y_k)p)$$

With our notation, we get:

$$\frac{1}{1-y_k}\epsilon_k^S S_k + \frac{\frac{\partial p}{\partial y_k}}{p}\Big(\sum_{i\in I}\epsilon_i^D D_i - \sum_{i\in I}\epsilon_i^S S_i\Big) = 0$$

From this we get:

$$\frac{\frac{\partial p}{\partial y_k}}{p} = \frac{1}{(1-y_k)}\frac{\epsilon_k^S \frac{S_k}{S}}{\epsilon^S - \epsilon^D}$$

$\square$

The formula just derived, $\frac{\frac{\partial p}{\partial x_k}}{p} = \frac{1}{(1+x_k)}\frac{\epsilon_k^D \frac{D_k}{D}}{\epsilon^S - \epsilon^D}$, can be explained intuitively as follows: an increase in player k's ad valorem tax rate x_k by one percentage point means an increase in the consumers' price of $\frac{1}{(1+x_k)}$ percent. This reduces player k's oil demand in proportion to its domestic price elasticity of demand, ϵ_k^D. To obtain the relative change in global oil demand that this constitutes, we have to multiply by its share of global oil consumption, $\frac{D_k}{D}$. Per percent reduction in global oil demand this translates into an adjustment in the global oil price of $\frac{1}{\epsilon^S - \epsilon^D}$ percent: the less responsive global demand and supply, the stronger the price adjustment required for market clearing to be reestablished.

The corresponding formula for the effect of increasing the ad valorem tax on oil extraction, $\frac{\frac{\partial p}{d y_k}}{p} = \frac{1}{(1-y_k)}\frac{\epsilon_k^S \frac{S_k}{S}}{\epsilon^S - \epsilon^D}$, is analogous. This is because all that matters for the price adjustment is how much at a given world market price global demand will differ from global supply.

Lemma 2.

$$\frac{\partial W_k}{\partial x_k} = pD_k\frac{x_k}{1+x_k}\epsilon_k^D + \frac{\partial p}{\partial x_k}(S_k - D_k + x_k D_k \epsilon_k^D + y_k S_k \epsilon_k^S)$$

Moreover, for $j \neq k$ we have:

$$\frac{\partial W_j}{\partial x_k} = \frac{\partial p}{\partial x_k}(S_j - D_j + x_j D_j \epsilon_j^D + y_j S_j \epsilon_j^S)$$

Also:

$$\frac{\partial W_k}{\partial y_k} = -pS_k\frac{y_k}{1-y_k}\epsilon_k^S + \frac{\partial p}{\partial y_k}(S_k - D_k + x_k D_k \epsilon_k^D + y_k S_k \epsilon_k^S)$$

Moreover, for $j \neq k$ we have:

$$\frac{\partial W_j}{\partial y_k} = \frac{\partial p}{\partial x_k}(S_j - D_j + x_j D_j \epsilon_j^D + y_j S_j \epsilon_j^S)$$

Proof. Differentiating the expression

$$W_k = \underbrace{\int_{v=(1+x_k)p}^{\infty} D_k(v)dv}_{\textbf{consumer surplus}} + \underbrace{px_k D_k((1+x_k)p) + py_k S_k((1-y_k)p)}_{\textbf{tax revenue}} + \underbrace{\int_{v=S_k^{-1}(0)}^{p(1-y_k)} S_k(v)dv}_{\textbf{producer surplus}}$$

with respect to x_k yields:

$$\frac{\partial W_k}{\partial x_k} = -pD_k((1+x_k)p) + pD_k((1+x_k)p) + p^2 x_k D_k'((1+x_k)p) + \frac{dp}{dx_k}(-(1+x_k)D_k + x_k D_k + (1+x_k)px_k D_k' + y_k S_k + (1-y_k)py_k S_k' +$$

$$\frac{\partial W_k}{\partial x_k} = pD_k\frac{x_k}{1+x_k}\epsilon_k^D + \frac{dp}{dx_k}(-D_k + (1+x_k)px_k D_k' + S_k + (1-y_k)py_k S_k')$$

$$\frac{\partial W_k}{\partial x_k} = pD_k\frac{x_k}{1+x_k}\epsilon_k^D + \frac{dp}{dx_k}(S_k - D_k + x_k D_k \epsilon_k^D + y_k S_k \epsilon_k^S)$$

Analogously, we get:

$$\frac{\partial W_k}{\partial y_k} = -pS_k \frac{y_k}{1-y_k}\epsilon_k^S + \frac{dp}{dy_k}(S_k - D_k + x_k D_k \epsilon_k^D + y_k S_k \epsilon_k^S)$$

□

Let us now interpret the expression for the demand side just derived:

$$\frac{\partial W_k}{\partial x_k} = pD_k \frac{x_k}{1+x_k}\epsilon_k^D + \frac{\partial p}{\partial x_k}(S_k - D_k + x_k D_k \epsilon_k^D + y_k S_k \epsilon_k^S)$$

by slightly rearranging it:

$$\begin{aligned} \frac{\partial W_k}{\partial x_k} = & \underbrace{x_k p}_{\substack{\textbf{demand}\\ \textbf{side}\\ \textbf{wedge}}} \underbrace{\underbrace{\left(\frac{1}{1+x_k} + \frac{\frac{\partial p}{\partial x_k}}{p}\right)}_{\substack{\textbf{induced net \%}\\ \textbf{change in}\\ \textbf{consumer price}}} \epsilon_k^D D_k}_{\textbf{change in oil consumption}} \\ & - \underbrace{\frac{\partial p}{\partial x_k}(D_k - S_k)}_{\textbf{terms of trade effect}} \\ & + \underbrace{y_k p}_{\substack{\textbf{supply}\\ \textbf{side}\\ \textbf{wedge}}} \underbrace{\underbrace{\frac{\frac{\partial p}{\partial x_k}}{p}}_{\substack{\textbf{induced net \%}\\ \textbf{change in}\\ \textbf{consumer price}}} \epsilon_k^S S_k}_{\textbf{change in oil production}} . \end{aligned}$$

Also, let us now interpret the expression for the supply side just derived:

$$\frac{\partial W_k}{\partial y_k} = -pS_k \frac{y_k}{1-y_k}\epsilon_k^S + \frac{\partial p}{\partial y_k}(S_k - D_k + x_k D_k \epsilon_k^D + y_k S_k \epsilon_k^S)$$

by slightly rearranging it:

$$\frac{\partial W_k}{\partial y_k} = \underbrace{y_k p}_{\substack{\textbf{supply}\\\textbf{side}\\\textbf{wedge}}} \underbrace{\underbrace{\left(\frac{-1}{1-y_k} + \frac{\frac{\partial p}{\partial x_k}}{p}\right)}_{\substack{\textbf{induced net \%}\\\textbf{change in}\\\textbf{producer price}}} \epsilon_k^D D_k}_{\textbf{change in oil consumption}}$$

$$- \underbrace{\frac{\partial p}{\partial x_k}(D_k - S_k)}_{\textbf{terms of trade effect}}$$

$$+ \underbrace{x_k p}_{\substack{\textbf{demand}\\\textbf{side}\\\textbf{wedge}}} \underbrace{\underbrace{\frac{\frac{\partial p}{\partial x_k}}{p}}_{\substack{\textbf{induced net \%}\\\textbf{change in}\\\textbf{consumer price}}} \epsilon_k^D D_k}_{\textbf{change in oil production}} .$$

Lemma 3. *The marginal change in climate damages induced by a change in the tax rate x_k can be expressed as:*

$\eta_k \frac{dS}{dx_k} = \eta_k S \frac{\frac{dp}{dx_k}}{p} \epsilon^S$

The marginal change in climate damages induced by a change in the tax rate y_k can be expressed as:

$\eta_k \frac{dD}{dy_k} = \eta_k D \frac{\frac{dp}{dy_k}}{p} \epsilon^D$

Proof. $\frac{d}{dx_k}(\eta_k \sum_{i \in I} S_i((1-y_i)p)) = \eta_k \frac{dp}{dx_k} \sum_{i \in I}(1-y_i)S_i'((1-y_i)p) = \eta_k \frac{\frac{dp}{dx_k}}{p} \sum_{i \in I} S_i \epsilon_i^S = \eta_k S \frac{\frac{dp}{dx_k}}{p} \epsilon^S$ □

Lemma 4. *Player k's best response satisfies the following:*

$x_k = \frac{\frac{D_k - S_k}{D} + \sum_{j \neq k}(-\alpha_{kj})(\frac{S_j - D_j}{D} + x_j \frac{D_j}{D}\epsilon_j^D + y_j \frac{S_j}{S}\epsilon_j^S) + \frac{\eta_k}{p}\frac{S_{-k}}{S}\epsilon_{-k}^S}{\frac{S_{-k}}{S}\epsilon_{-k}^S - \frac{D_{-k}}{D}\epsilon_{-k}^D}$

$y_k = -\frac{\frac{D_k - S_k}{D} + \sum_{j \neq k}(-\alpha_{kj})(\frac{S_j - D_j}{D} + x_j \frac{D_j}{D}\epsilon_j^D + y_j \frac{S_j}{S}\epsilon_j^S) + \frac{\eta_k}{p}\frac{D_{-k}}{D}\epsilon_{-k}^D}{\frac{S_{-k}}{S}\epsilon_{-k}^S - \frac{D_{-k}}{D}\epsilon_{-k}^D}$

Proof. Differentiating $U_k = W_k + \sum_{i \neq k} \alpha_{ki} W_i - \underbrace{\eta_k S}_{\textbf{climate damages}}$ with respect to x_k yields as the first order condition for k to play a best response:

$0 = \frac{\partial U_k}{\partial x_k} = \frac{\partial W_k}{\partial x_k} + \sum_{i \neq k} \alpha_{ki} \frac{\partial W_i}{\partial x_k} - \eta_k \frac{dS}{dx_k}$

Now using Lemmas 2 and 3 yields:

$0 = pD_k \frac{x_k}{1+x_k}\epsilon_k^D + \frac{dp}{dx_k}(S_k - D_k + x_k D_k \epsilon_k^D + y_k S_k \epsilon_k^S) + \sum_{i \neq k} \alpha_{ki}(\frac{dp}{dx_k}(S_j - D_j + x_j D_j \epsilon_j^D + y_j S_j \epsilon_j^S)) - \eta_k S \frac{\frac{dp}{dx_k}}{p}\epsilon^S$

Dividing by our result from Lemma 1, $\frac{\partial p}{dx_k} = \frac{p}{(1+x_k)} \frac{\epsilon_k^D \frac{D_k}{D}}{\epsilon^S - \epsilon^D}$, yields:

$0 = x_k D(\epsilon^S - \epsilon^D) + (S_k - D_k + x_k D_k \epsilon_k^D + y_k S_k \epsilon_k^S) + \sum_{i \neq k} \alpha_{ki}(S_j - D_j + x_j D_j \epsilon_j^D + y_j S_j \epsilon_j^S) - \frac{\eta_k}{p} S \epsilon^S$

We can view this as a linear equation in (x_k, y_k) and rearrange this as:

$x_k(\epsilon^S - (\epsilon^D - \frac{D_k}{D}\epsilon_k^D)) + y_k \frac{S_k}{S}\epsilon_k^S = \frac{D_k - S_k}{D} + \sum_{j \neq k} \alpha_{ki}(\frac{D_j - S_j}{D} - x_j \frac{D_j}{D}\epsilon_j^D - y_j \frac{S_j}{S}\epsilon_j^S) + \frac{\eta_k}{p}\epsilon^S$

Analogously, one can derive:

$x_k \frac{D_k}{D}\epsilon_k^D + y_k(\epsilon^D - (\epsilon^S - \frac{S_k}{S}\epsilon_k^S)) = \frac{D_k - S_k}{D} + \sum_{j \neq k} \alpha_{ki}(\frac{D_j - S_j}{D} - x_j \frac{D_j}{D}\epsilon_j^D - y_j \frac{S_j}{S}\epsilon_j^S) + \frac{\eta_k}{p}\epsilon^D$

Substracting these two equations yields:

$(x_k + y_k)(\epsilon^S - \epsilon^D) = \frac{\eta_k}{p}(\epsilon^S - \epsilon^D)$

Hence:

$x_k + y_k = \frac{\eta_k}{p}$

Substituting $y_k = \frac{\eta_k}{p}\epsilon^S - x_k$ into the optimality condition for x_k yields:

$$x_k(\epsilon^S - (\epsilon^D - \tfrac{D_k}{D}\epsilon^D_k)) + (\tfrac{\eta_k}{p} - x_k)\tfrac{S_k}{S}\epsilon^S_k = \tfrac{D_k - S_k}{D} + \textstyle\sum_{j\neq k}\alpha_{ki}(\tfrac{D_j - S_j}{D} - x_j\tfrac{D_j}{D}\epsilon^D_j - y_j\tfrac{S_j}{S}\epsilon^S_j) + \tfrac{\eta_k}{p}\epsilon^S$$
$$x_k(\tfrac{S_{-k}}{S}\epsilon^S_{-k} - \tfrac{D_{-k}}{D}\epsilon^D_{-k}) = \tfrac{D_k - S_k}{D} + \textstyle\sum_{j\neq k}\alpha_{ki}(\tfrac{D_j - S_j}{D} - x_j\tfrac{D_j}{D}\epsilon^D_j - y_j\tfrac{S_j}{S}\epsilon^S_j) + \tfrac{\eta_k}{p}\tfrac{S_{-k}}{S}\epsilon^S_{-k}$$
$$x_k = \frac{\frac{D_k - S_k}{D} + \sum_{j\neq k}(-\alpha_{ki})(\frac{S_j - D_j}{D} + x_j\frac{D_j}{D}\epsilon^D_j + y_j\frac{S_j}{S}\epsilon^S_j) + \frac{\eta_k}{p}\frac{S_{-k}}{S}\epsilon^S_{-k}}{\frac{S_{-k}}{S}\epsilon^S_{-k} - \frac{D_{-k}}{D}\epsilon^D_{-k}}$$
$$-y_k = \frac{\frac{D_k - S_k}{D} + \sum_{j\neq k}(-\alpha_{ki})(\frac{S_j - D_j}{D} + x_j\frac{D_j}{D}\epsilon^D_j + y_j\frac{S_j}{S}\epsilon^S_j) + \frac{\eta_k}{p}\frac{D_{-k}}{D}\epsilon^D_{-k}}{\frac{S_{-k}}{S}\epsilon^S_{-k} - \frac{D_{-k}}{D}\epsilon^D_{-k}} \qquad \square$$

Corollary 1. ***("Pigou despite leakage".)*** *Player k's best response always satisfies the following:* $(x_k + y_k)p = \eta_k$

This corollary states that regardless of trade flows, elasticities and resulting leakage rates, it is always optimal for a government to respect the following version of the Pigouvian injunction: "set the taxes such that *their sum* equals the external cost". If the player k makes up the entire world then respecting this injunction is sufficient for optimality and it is immaterial how the taxes are split between consumption and production. However, in the case where the player k is only a part of the world then there is generically a unique combination, pinned down in Lemma 4.

Now let us interpret the formula for the optimal consumption tax rate. For concreteness, let us consider the EU and let us suppose that the only non-zero $\alpha_{EU,j}$ is for $j = Russia$. Then we get:

$$x_{EU} = \frac{1}{\frac{S_{-EU}}{S}\epsilon^S_{-EU} - \frac{D_{-EU}}{D}\epsilon^D_{-EU}}\Bigg(\underbrace{\underbrace{\frac{D_{EU} - S_{EU}}{D}}_{\substack{\textbf{lower EU}\\ \textbf{expenditure for}\\ \textbf{the net imports}}}}_{\textbf{terms of trade effect}} \underbrace{-\alpha_{EU,Russia}(\underbrace{\frac{S_{Russia} - D_{Russia}}{D}}_{\substack{\textbf{lower Russian}\\ \textbf{revenue from}\\ \textbf{the net exports}}} + \underbrace{x_{Russia}\frac{D_{Russia}}{D}\epsilon^D_{Russia}}_{\substack{\textbf{change in Russia's}\\ \textbf{deadweight loss}\\ \textbf{on the consumption side}}} + \underbrace{y_{Russia}\frac{S_{Russia}}{S}\epsilon^S_{Russia}}_{\substack{\textbf{change in Russia's}\\ \textbf{deadweight loss}\\ \textbf{on the consumption side}}})}_{\substack{\textbf{geopolitical}\\ \textbf{externality}}} + \frac{\eta_{EU}}{p}\frac{S_{-EU}}{S}\epsilon^S_{-EU}\Bigg)$$

Thus we can write the EU's best response profile as follows:
$x_{EU} = \text{terms of trade effect} + \text{geopolitical externality} + \lambda\frac{\eta_{EU}}{p}$
$y_{EU} = -\text{terms of trade effect} - \text{geopolitical externality} + (1 - \lambda)\frac{\eta_{EU}}{p}$
where $\lambda := \frac{\frac{S_{-EU}}{S}\epsilon^S_{-EU}}{\frac{S_{-EU}}{S}\epsilon^S_{-EU} - \frac{D_{-EU}}{D}\epsilon^D_{-EU}}$.

Now let us look more in detail at the geopolitical externality. Recall that $\alpha_{EU,Russia}$ is negative since the Russian aggression harms the EU. The term $\underbrace{x_{Russia}\frac{D_{Russia}}{D}\epsilon^D_{Russia}}_{\substack{\textbf{change in}\\ \textbf{Russia's deadweight}\\ \textbf{on the consumption side}}}$ can be interpreted as follows for the current situation where $x_{Russia} < 0$, i.e. where the Russian government is effectively subsidizing their population's oil consumption (Note that the appropriate concept for the effective ad valorem tax rate x_{Russia} corresponds to the difference between the consumer price and the world market price + distribution cost.): A side effect of an increase in x_{EU} is an increase in oil consumption in Russia due to the induced decrease in the world market price. Given that Russia subsidizes domestic oil consumption, this means an increase in the deadweight loss for Russia which explains why the term $x_{Russia}\frac{D_{Russia}}{D}\epsilon^D_{Russia} > 0$ contributes positively in the expression for the EU's optimal tax rate x_{EU}: From the EU's perspective it is to be welcomed that Russia is distorting its own economy by subsidizing oil consumption and indirectly the cost of this distortion gets increased via increases in x_{EU}.

Note that our formula characterizes the EU's best response *in general*, whether or not Russia's policy is itself a best response or not. However, note that the negative x_{Russia} that we observe might in fact be consistent with Russia playing a best response, given its low (or possibly even negative) national social cost of carbon and its net oil exports. Importantly, therefore, the effect unveiled in the above formula and discussed in the preceding paragraph, namely that the deadweight losses accruing to Russia from its tax policy provide a further effect that drives up the EU's optimal oil consumption tax, does not require there to be any irrationality on the part of Russia's tax policy. In fact, it is fully consistent with Russia itself playing a best response.

2.2 Calibration

2.2.1 Approximate calibration excluding the deadweight loss effects in the geopolitical externality term

For this, results for each fossil fuel can be found here, drawing on Beaufils et al. 2025.

We get for oil:
$\frac{D_{EU}-S_{EU}}{D} = 0.12$
$\alpha_{EU,Russia} = 2.9$
$\frac{S_{Russia}-D_{Russia}}{D} = 0.08$
$\frac{\eta_{EU}}{p}\frac{S_{-EU}}{S}\epsilon^S_{-EU} = \frac{33}{186}0.990.13 = 0.022$
$\alpha_{EU,Russia}\frac{S_{Russia}-D_{Russia}}{D} = 2.90.08 = 0.23$

With this we get an optimal (from a self-interested perspective) ad valorem tax rate on oil consumption for the EU of $x_{EU} = 0.64$ with the following decomposition:
climate benefit: 0.04
terms of trade benefit: 0.20
geopolitical benefit: 0.40

Also we get an optimal (from a self-interested perspective) ad valorem tax rate on oil production for the EU of $y_{EU} = 0.20 + 0.40 - 0.15 = 0.45$ with the following decomposition:
climate benefit: 0.15
terms of trade cost: 0.20
geopolitical cost: 0.40

2.2.2 Adjustments implied by including the deadweight loss effects in the geopolitical externality term

To calibrate y_{Russia}, we need to estimate the "supply side wedge", i.e. the hypothetical ad valorem rate on Russian oil extraction that would lead a firm that takes world market prices as given and maximises profits to implement the current allocation on the supply side. Given that currently Russia seems to be at a corner solution where they extract as much as they can with the available technology, we have $y_{Russia} = 0$.

The ad valorem tax rate on consumption is $x_{Russia} = -0.5$.
$D_{Russia} = 3.63$
$S_{Russia} = 10.75$
$\epsilon^D_{Russia} = -0.45$ (assuming the elasticity is the same as the global one)
$\Longrightarrow$ Taking into account the deadweight loss effects increases the geopolitical externality by the proportion $x_{Russia}\epsilon^D_{Russia}\frac{D_{Russia}}{S_{Russia}-D_{Russia}} = 0.11$

2.3 Discussion of modifications of the model

Let us now consider how the results would change if we were to change the timing in the game. Specifically, let us first consider the two-stage game where the EU moves first and Russia moves second. For simplicity, let us consider the special case where $\eta_{Russia} = 0$ and $U_{Russia} = W_{Russia}$. This assumption can be justified as a decent approximation based on the observation that the EU's spending on supporting Ukraine is a sufficiently small part of their GDP that $\alpha_{Russia,EU}$ is small (Spiro ...). For that case, it can be shown via the envelope theorem that $\frac{\partial W_{Russia}}{\partial x_{EU}}$ is exactly as computed above, where we assumed that Russia does not react to the EU's choice of x_{EU}. The reason is as follows: the choice of (x_{EU}, y_{EU}) in the first stage of the game can be viewed as changing parameters that enter Russia's maximisation problem in the second stage of the game. The envelope theorem tells us that the change in W_{Russia} can be computed by naively ignoring Russia's re-optimization.

It follows that at the subgame perfect equilibrium, the EU's choice of (x_{EU}, y_{EU}) will only differ from the EU's static best response (i.e. what we have computed above) to the extent that the the changes in (x_{Russia}, y_{Russia}) induced in the second stage affect p. But under the assumptions we have now made, we have:

$$x_{Russia} = \frac{\frac{D_{Russia} - S_{Russia}}{D}}{\frac{S_{-Russia}}{S}\epsilon^S_{-Russia} - \frac{D_{-Russia}}{D}\epsilon^D_{-Russia}}$$

$$y_k = -\frac{\frac{D_{Russia} - S_{Russia}}{D}}{\frac{S_{-Russia}}{S}\epsilon^S_{-Russia} - \frac{D_{-Russia}}{D}\epsilon^D_{-Russia}}$$

It is plausible that these values will not be greatly changed by changes in (x_{EU}, y_{EU}). Thus the resulting induced change in p is likely to be relatively small. This suggest that the EU's (x_{EU}, y_{EU}) at the Subgame Perfect Nash Equilibrium is similar to its choices at the Nash Equilibrium in the simultaneous move game, as computed above.

Now let us informally venture a further step outside of the above model. It is plausible that governments assign some value to not having consumer prices for oil go too high, as suggested by commonly observed decreases in fuel taxes in times of high world market prices. In that case the forward looking logic encapsulated in the SPNE yields a further effect in favor of a high x_{EU} out of self-interst for the EU: by lowering the world market price p (via a high x_{EU}), the EU will cause other countries to increase their x_i, thereby further decreasing p and thereby benefiting the EU directly via lower oil import expenses and indirectly via lower oil revenues accruing to Russia.

3 Empirical calibration for oil

3.1 Global annual expenditure on oil

$2296\,billion$ for the global annual dollar amount of spending on oil [1]

3.2 Elasticities

$e_{d\,oil} = 0.5$, based on Keen et al. (2019) for the price elasticity of oil demand

$e_{s\,oil} = 0.32$, based on Golombek et al. (2018) for the price elasticity of oil supply

3.3 Social cost of carbon

$\eta = 0.24$ based on a social cost of carbon of \$36 per ton of CO2 (based on EPA (2015)).[2]

3.4 Oil trade

Here are the countries with the largest oil exports and imports:[3]

[1] World oil consumption in 2019 was 98.27M bbl/d. The eia states: The price of Brent crude oil, the international benchmark, averaged \$64 per barrel (b) in 2019. This gives that global oil consumption is \$2.296 trillion per year (in 2019).

[2] Recall that by Lemma **??** our demand and supply specifications have the following implicit normalisation: Price at the status quo is 1. Thus η is the ratio of the social cost of a unit of oil divided by its price. To compute this we use the following: The EPA states: The average carbon dioxide coefficient of distillate fuel oil is 429.61 kg CO2 per 42-gallon barrel (EPA 2018). The fraction oxidized to CO2 is 100 percent (IPCC 2006). The eia states: The price of Brent crude oil, the international benchmark, averaged \$64 per barrel (b) in 2019. Assuming a social cost of carbon of \$36 per ton of CO2, we get: $\eta/p = 36 * 0.42961/64$=0.24.

[3] According to https://www.worldstopexports.com/worlds-top-oil-exports-country/, 43.8% of all oil is traded.The shares of this trade corresponding to the countries is recorded on https://www.worldstopexports.com/coal-exports-country/.

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4 Numerical results for oil

5 Conclusion and limitations

References

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